

Frontier Research in Generative AI & NLP

MSc Thesis Topic Selection: A Review of 9 Research Directions

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Research Methodology & Ranking Framework



Literature Survey

Spanning 2020–2026 across RAG, Quantization, and Bangla NLP.



Identification

Extracted 9 viable niches within the current GenAI space.



Scoring

Ranked via 2x2 matrix: Novelty vs. MSc Feasibility.



Shortlisting

Selected Top 3 based on departmental alignment.

Criterion	Core Question for Evaluation	Success Metric
MSc Feasibility	Can this be completed in 12 months with BUET compute?	GPU utilization / Time
Dataset Availability	Does high-quality training data exist for Bangla?	Token Count / Cleanliness
Research Novelty	Is there a genuine gap in 2026 frontier literature?	SOTA Comparison Gap
Personal Alignment	Does it match my SE + GenAI background?	Skill Synergy

Topic 1: RAG Optimization ⭐ (Top Pick)

Primary Research Direction

⚠️ Problem & Gap

Permutation Sensitivity: Document ordering causes reasoning failures in standard RAG pipelines.

- Bangla RAG systems are largely absent in current literature.
- Morphologically complex languages face unique chunking issues.
- Gap: Hallucination amplification in multi-document retrieval.

🎯 Proposed Thesis Angle

"Adaptive chunking and hybrid retrieval optimization for Bangla QA using RAG"

Metric: RC
References Completeness

Metric: BI
Block Integrity

Metric: ICC
Intrachunk Cohesion

Metric: SC
Size Compliance

2026 Frontier Frameworks

Stable-RAG: Hidden-state clustering during decoding to stabilize reasoning.

VOTE-RAG: Multi-agent retrieval voting + response consensus.

Hybrid Retrieval: BM25 sparse + dense (FAISS) via RRF fusion.

Topic 2: Hallucination Detection & Mitigation

Rank 2: Systems Optimization

Factuality

Contradicts real-world knowledge. Detected via citation analysis.

Faithfulness

Contradicts provided context. Detected via reasoning graphs.

Joint Optimization

Embedding detection heads inside the model itself (2026 SOTA).

 The RAGognizer Architecture (2026)

Proposed thesis: "Joint optimization for hallucination-aware generation in Bangla LLMs"

Component	3-layer MLP detection head inserted directly inside LLM layers	Outcome	92.6% AUROC on Qwen3-4B
Loss Function	$L = L_{CE} + \lambda L_{BCE}$ (Joint Cross-Entropy and Binary CE)	Signature	Attention sink analysis for citation validity (FACTUM)

Topic 3: Multilingual & Low-Resource Bangla

Rank 3: Regional Impact

TituLLMs & Sangraha (2026)

37B Tokens

- **Bangla MMLU:** 87,869 entries (BCS/Univ).
- **SOMAJGYAAN:** 4,234 Social Knowledge questions.
- **Problem:** Tokenizer efficiency (TPW) in earlier models (BongLLaMA).

Domain-Specific Finetuning

Focus on Bangla Legal/Medical QA.

- **EST Pipeline:** Expressive Semantic Translation for synthetic data.
- **BanglaMedQA:** 2,000 medical QA entries.
- **Gap:** Overcoming the "Embedding Barrier" for native scripts.

Note: Aligning with BUET's **BanglaBERT** tradition to ensure continuity in regional research leadership.

Topic 4: Efficient & Lightweight LLMs

Proposed Angle: "Quality-efficiency trade-offs in sub-1-bit quantized LLMs for Bangla tasks."

Precision Level	70B Model Size	Hardware Requirement	2026 Frontier Tech
FP16	138 GB	Enterprise GPU Cluster (H100)	Standard Baseline
4-bit (QLoRA)	~35 GB	High-end Consumer (RTX 4090)	Dettmers et al. (2023)
1-bit (BTC-LLM)	~10 GB	Mid-range Consumer (RTX 3060)	Binary Pattern Clustering
Sub-1-bit (NanoQuant)	5.35 GB	Standard Consumer (8GB GPU)	ADMM Binary Factorization

Feasibility Note: Sub-1-bit models now achieve 20.11 tokens/second on standard 8GB hardware. Ideal for researchers with limited compute access.

Topic 5: Code Generation & Software Engineering

The "Solved" Fallacy

HumanEval and MBPP are increasingly contaminated by training data. 2026 requires real-world repo evaluation.

Waterfall Degradation

Agents understand high-level intent but fail during concrete execution planning in large repos.

RACE-bench (2026)

Dual-track evaluation: Patch correctness + Intermediate reasoning quality.

CodeScaleBench-Org Findings

Evaluation on 220 enterprise tasks (repos > 5GB). MCP-augmented agents achieved significant recall gains:

0.277
MCP Agent Recall

0.127
Baseline Agent Recall

Topic 6: Evaluation Frameworks for Generative Models

Standard Bias in "LLM-as-Judge"

Position Bias: Favors the first response.

Verbosity Bias: Longer = Higher score.

Self-Preference: Favors same-family models.

★ Prometheus (2026)

Specialized judge model. High alignment with human judgment.

0.897

Correlation with human evaluators.

Case Study: Bangla Legal Findings (2026)

LLM judges missed fabricated citations identified by human Bangladeshi lawyers. **Conclusion:** Human-in-the-loop remains critical for safety-critical Bangla domains.

Topic 7: Fine-Tuning vs. Prompt Engineering

The "BetterTogether" Hybrid Strategy (2026): Reframing the binary choice into a 3-stage optimization pipeline.

Stage 1

DSPy Optimization

Programmatic prompt optimization; treating prompts as trainable parameters.

Stage 2

SFT (Supervised)

"Baking in" formatting and domain-specific stylistic requirements.

Stage 3

GRPO (RL)

Group Relative Policy Optimization to push the reasoning ceiling.

14x

Cost-Efficiency Gain

This hybrid approach allows small models (7B) to outperform large frontier models (GPT-5) on repetitive, domain-specific Bangla tasks.

Topic 8: AI Agents & Tool-Using LLMs

Planning Loops

ReAct: Thought → Action → Observation cycles.

Reflexion: Verbal reinforcement learning via self-critique.

MCP Protocol (2026)

Model Context Protocol: Standard for enterprise agent tool-use.
Enables cross-repo symbol resolution and semantic search.

Thesis Focus: "MCP-augmented Bangla Academic Agent"

Restricting domain to academic research assistance to maintain thesis scope while testing 2026 tool-using capabilities in a Bangla context.

Topic 9: Ethics & Safety of GenAI

IndicSafe 2026: Safety Drift Findings

Model	Safe Response %	Unsafe %	Refusal %
Grok-3	83.6%	0.9%	12.4%
LLaMA 4	78.7%	5.2%	12.2%
GPT-4o Mini	58.5%	20.7%	16.1%
Qwen 1.5	4.5%	54.5%	16.0%

Critical Gap: "Safety Drift" shows only 12.8% cross-language agreement. A model safe in English often generates toxic or biased content in Bangla due to lack of cultural grounding.

Recommendation & Final Shortlist

Selection 1

RAG Optimization

Strongest for technical systems contribution.
Concrete benchmarks (Stable-RAG) available.

Selection 2

Hallucination Detect

Natural secondary contribution to Selection 1.
High publishable novelty for Bangla.

Selection 3

Bangla Domain-FT

Highest regional impact. Aligns with BUET's BanglaBERT research legacy.

Decision Matrix for Supervisor

- ✓ **Computing:** Low-compute topics (4, 7, 9) vs High-compute (3, 5).
- ✓ **Output:** Single system (1, 2, 8) vs Comparative study (6, 7, 9).

- ✓ **Data:** Synthetic-heavy (7, 10) vs Real-data (3, 11).

Thank You!

Open for Guidance & Discussion

Current Goal: Finalize 1 direction today.

Next Step: Detailed proposal in 14 days.

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